

Assessment - 2

eg. If consider a website that user enter the age to 18 to 50 then eligible for getting license otherwise not eligible. using the scenario write the test cases for boundary value analysis.

TC-ID	Test scenario	test case	Pre condition	test steps	test data	expected result	Post Condition	Actual result	Status
TC-01	Age eligibility check	enter age = 18	website opened	1. enter 18 2. click submit	18	user is eligible	eligibility message displayed	user is eligible	Pass
TC-02	Age eligibility check	enter age = 19	website opened	1. enter 19 2. click submit	19	user is eligible	eligibility message displayed	user is eligible	Pass
TC-03	Age eligibility check	enter age = 34	website opened	1. enter 34 2. click submit	34	user is eligible	eligibility message displayed	user is eligible	Pass
TC-04	Age eligibility check	enter age = 49	website opened	1. enter 49 2. click submit	49	user is eligible	eligibility message displayed	user is eligible	Pass
TC-05	Age eligibility check	enter age = 50	website opened	1. enter 50 2. click submit	50	user is eligible	eligibility message displayed	user is eligible	Pass

TC-ID	Test scenario	Test case	Pre condition	Test steps	test data	Expected result	Post condition	Actual result	Status
TC-06	Age eligibility check	enter age = 17	website opened	1. enter 17 2. click submit	17	user not eligible	not eligible message displayed	user not eligible	fail
TC-07	age eligibility check	enter age = 51	website opened	1. enter 51 2. click submit	51	user not eligible	not eligible message displayed	user not eligible	fail
TC-08	age eligibility check	enter age = 0	website opened	1. enter 0 2. click submit	0	error message displayed	Input expected	error message displayed	fail
TC-09	age eligibility check	enter age = 100	website opened	1. enter 100 2. click submit	100	user not eligible	not eligible message displayed	user not eligible	fail
TC-10	age eligibility check	enter age = 1	website opened	1. enter -1 2. click submit	-1	error message displayed	Input expected	error message displayed	fail

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Q-2. A program reads 3 integers from the user, find which one is greatest and which one is smallest using equivalence partitioning.

TC-ID	Test Scenario	Test Case	Pre Condition	Test Steps	Test data			Expected result	Post condition	Actual result	Status.
					A	B	C				
TC-01	find the greatest & smallest	A = 10, B = 20, C = 30	Program is running	1. enter 10 2. enter 20 3. enter 30	10	20	30	greatest = 30 smallest = 10	Output displayed	correct result shown	Pass
TC-02	find the greatest & smallest	A = 15, B = 15, C = 5	Program is running	1. enter 15 2. enter 15 3. enter 5	15	15	5	greatest = 15 smallest = 5	Output displayed	correct result shown	Pass
TC-03	find the greatest & smallest	A = 7, B = 7, C = 7	Program is running	1. enter 7 2. enter 7 3. enter 7	7	7	7	greatest = 7 smallest = 7	Output displayed	correct result shown	Pass
TC-04	find the greatest & smallest	A = -5 B = -10 C = -2	Program is running	1. enter -5 2. enter -10 3. enter -2	-5	-10	-2	greatest = -2 smallest = -10	Output displayed	correct result shown	Pass
TC-05	find the greatest & smallest	A = 0 B = -3 C = 9	Program is running	1. enter 0 2. enter -3 3. enter 9	0	-3	9	greatest = 9 smallest = -3	Output displayed	correct result shown	Pass

TC-ID	Test scenario	Test case	Pre condition	Test steps	test data	expected result	Post condition	Actual result	Status
					A B C				
TC-06	find the greatest & smallest	A = a A = 10, B = 5 C = 10	Program is running	1. enter a 2. enter b 3. enter c	a 5 10	error message	Input rejected	error message	fail
TC-07	find the greatest & smallest	A = 10, B = , C = 5	Program is running	1. enter 10 2. enter - 3. enter 5	10 5	error message	Input rejected	error message	fail
TC-08	find the greatest & smallest	A = 100, B = 50, C = -50	Program is running	1. enter 100 2. enter 50 3. enter -50	100 50 -50	greatest = 100 smallest = -50	output displayed	correct result shown	Pass
TC-09	find the greatest & smallest	A = 1, B = 0, C = 0	Program is running	1. enter 1 2. enter 0 3. enter 0	1 0 0	greatest = 1 smallest = 0	Output displayed	correct result shown	Pass
TC-10	find the greatest & smallest	A = -1 B = -2 C = -3	Program is running	1. enter -1 2. enter -2 3. enter -3	-1 -2 -3	greatest = -1 smallest = -3	Output displayed	correct answer shown	Pass

eg. A program takes an angle as input within the range 0 to 360 and determines in which quadrant the angle lies. design test cases using equivalent partitioning method.

TC-ID	Test Scenario	Test Case	Pre condition	test steps	test data	expected result	Post condition	Actual result	Status
TC-01	Identifies the quadrant	$A = 45^\circ$	Program is running	1. Run program 2. enter angle	45°	first quadrant	displays quadrant name	correct result shown	Pass
TC-02	Identifies the quadrant	$A = 120^\circ$	Program is running	1. Run program 2. enter angle	120°	second quadrant	displays quadrant name	correct result shown	Pass
TC-03	Identifies the quadrant	$A = 210^\circ$	Program is running	1. Run program 2. enter angle	210°	third quadrant	displays quadrant name	correct result shown	Pass
TC-04	Identifies the quadrant	$A = 300^\circ$	Program is running	1. Run program 2. enter angle	300°	fourth quadrant	displays quadrant name	correct result shown	Pass
TC-05	Identifies the quadrant	$A = 90^\circ$	Program is running	1. Run program 2. enter angle	90°	lies on axis	display quadrant name	correct result shown	Pass
TC-06	Identifies the quadrant	$A = 0^\circ$	Program is running	1. Run program 2. enter angle	0°	lies on axis	displays quadrant name	correct result shown	Pass

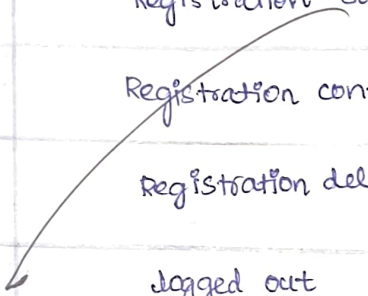
TC-ID	test scenario	test case	Pre condition	Test steps	Test data	Expected result	Post condition	Actual result	Status.
TC-07	Identifies the quadrant	A = 360°	Program is running	1. Run program 2. enter angle	360°	Lies on axis	displays quadrant name	correct result shown	Pass.
TC-08	Identifies the quadrant	A = -20°	Program is running	1. Run program 2. enter angle	-20°	Invalid input	Input rejected	error message shown	fail.
TC-09	Identifies the quadrant	A = 400°	Program is running	1. Run program 2. enter angle	400°	Invalid input	Input rejected	error message shown	fail
TC-10	Identifies the quadrant	A = ABC	Program is running	1. Run program 2. enter angle	ABC	Invalid input	Input rejected	error message shown	fail

Signature

10. To perform state table testing for student course registration system.

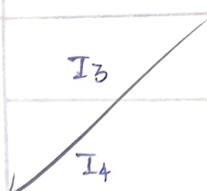
Step-1: Identify states

State ID	State Name
S0	Idle / Not logged in
S1	logged in
S2	viewing course list
S3	course selected
S4	Registration submitted
S5	Registration confirmed
S6	Registration deleted
S7	logged out



Step-2: Identify Inputs

Input-ID	Input
I1	Enter your valid credential
I2	Enter your Invalid credential
I3	view your list
I4	select courses
I5	submit registration
I6	Registration success
I7	Registration failed
I8	Logout



Step-3 : State Table

Current State	I/P	Next State	output
S0	I1	S1	login success
S0	I2	S0	login failed
S1	I3	S2	course 1 st displayed
S2	I4	S3	course selected
S3	I5	S4	registration submitted
S4	I6	S5	registration confirmed
S4	I7	S6	registration rejected
S5	I8	S7	logged out
S6	I8	S7	logged out
S1	I8	S7	logged out
S2	I8	S7	logged out
S3	I8	S7	logged out
S4	I8	S7	logged out

Step-4 : Test cases derived from the state table.

TC-ID	test scenario	test case	pre-condition	test steps	test data	expected result	Post condition	Actual result	Status
TC-01	login success	login with valid credentials	user on login page (s0)	1. enter name 2. enter password 3. login	< valid credential >	"login success"	state "s1"	"login success"	Pass.
TC-02	Invalid login	login with invalid credentials	user on login page (s0)	1. enter "name" 2. enter password 3. login	< invalid credential >	"login failed"	state "s0"	"login failed"	Pass fail
TC-03	view course list	display available courses	logged in (s1)	click view courses	-	course list displayed	state "s2"	course list displayed	Pass.
TC-04.	select courses	select course from list	course list opens (s2)	1. select course 2. click add	< course ID >	course select displayed successful	state "s3"	course select displayed successful	Pass
TC-05	submit registration	submit selected course	course selected (s3)	1. click submit 2. click add	< course - ID >	registration submitted message	state "s4"	registration submitted message	Pass
TC-06	registration successful	submitted confirmed registration	confirm registration (s4)	system checks seats	< seats allocated >	registration confirmed displayed	state "s5"	registration confirmed displayed	Pass.

TC-ID	Test scenario	Test case	Pre condition	Test steps	Test data	Expected result	Post condition	Actual result	Status
TC-07	registration failed	registration rejected	registration submitted (S4)	system checks seats	< seat filled >	"registration failed displayed"	state "S6"	"registration failed displayed"	fail
TC-08	logout	logged out from the system	Any logged state (S1/S2/S3/S4/S5/S6)	click logout button	—	"logout success"	state "S7"	"logout success"	Pass

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